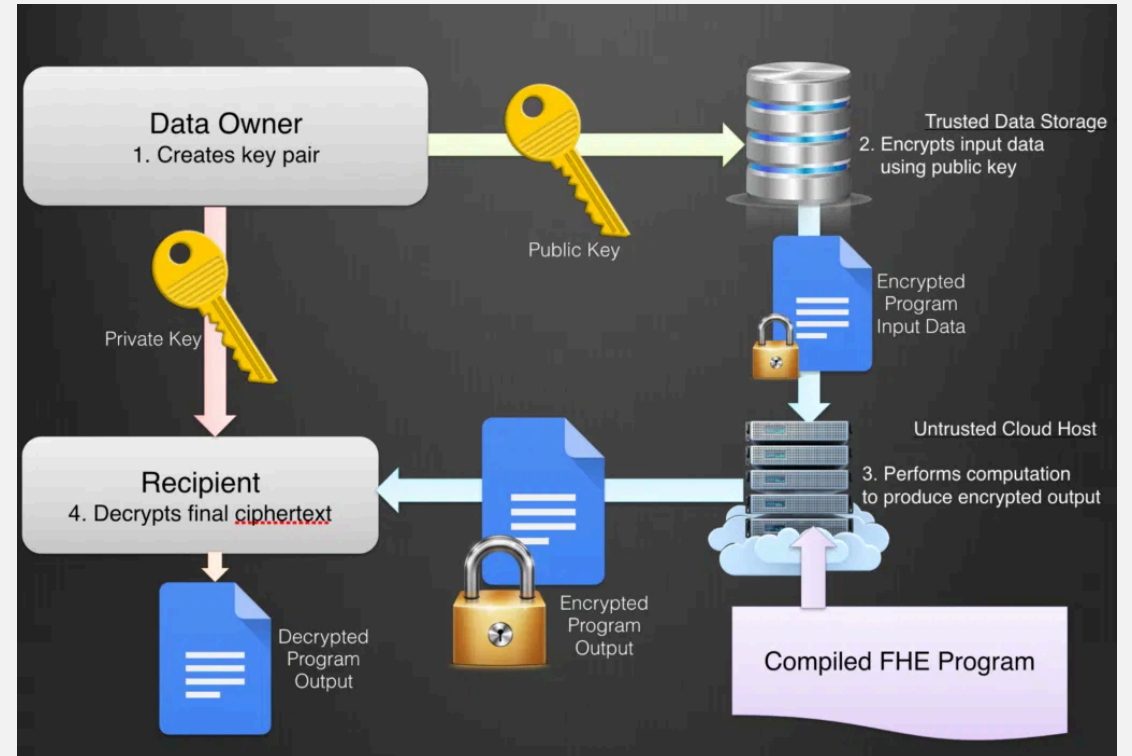


MANA FHE: Revolutionizing Privacy-Preserving Computation

- ▶ The 60-Year RNS Division Problem Solved with formal verification and exact arithmetic
- ▶ Unprecedented Performance orders of magnitude faster than SEAL, OpenFHE, and TFHE-rs
- ▶ Bootstrap-Free Rescaling enabling real-time privacy-preserving computation



K-Elimination: Constant-Time Exact Division

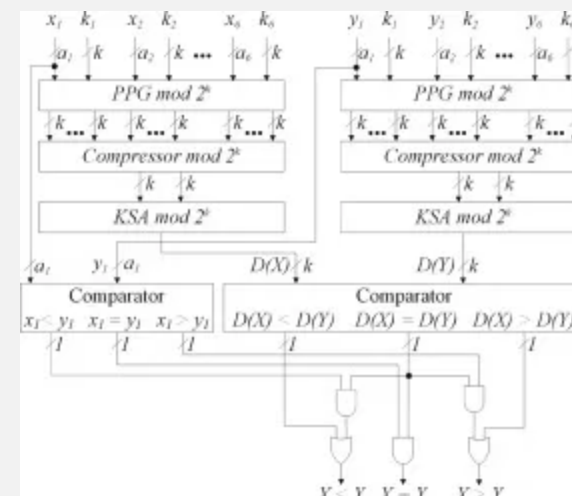
Traditional RNS division requires expensive $O(k^2)$ tracking or inexact floating-point estimation. K-Elimination solves this with a closed-form formula.

THE K-ELIMINATION FORMULA

$$k \equiv (vA - vM) \cdot M^{-1} \bmod A$$

Where vA is the anchor residue, vM is the main residue, and M^{-1} is the modular inverse. The phase differential between main and anchor residue spaces encodes the overflow count exactly.

- ✓ 100% exact recovery in $O(k)$ time with no explicit tracking required
- ✓ Enables bootstrap-free rescaling, eliminating the most expensive FHE operation
- ✓ Formally verified in Lean 4 and Coq with zero unproven statements



Dual-Track Exact Arithmetic: Zero-Drift

TRADITIONAL CKKS (APPROXIMATE)

Rounding errors accumulate with each multiplication. After 1000 operations, precision is significantly degraded. Unsuitable for deep circuits.

QMNF DUAL-TRACK (EXACT)

Maintains 100% exactness throughout computation. The 1000th operation is as precise as the first. Enables arbitrarily deep circuits.

- Eliminates approximation error accumulation (drift)
- Enables deep circuits without precision loss
- Essential for critical applications: medical, financial, scientific

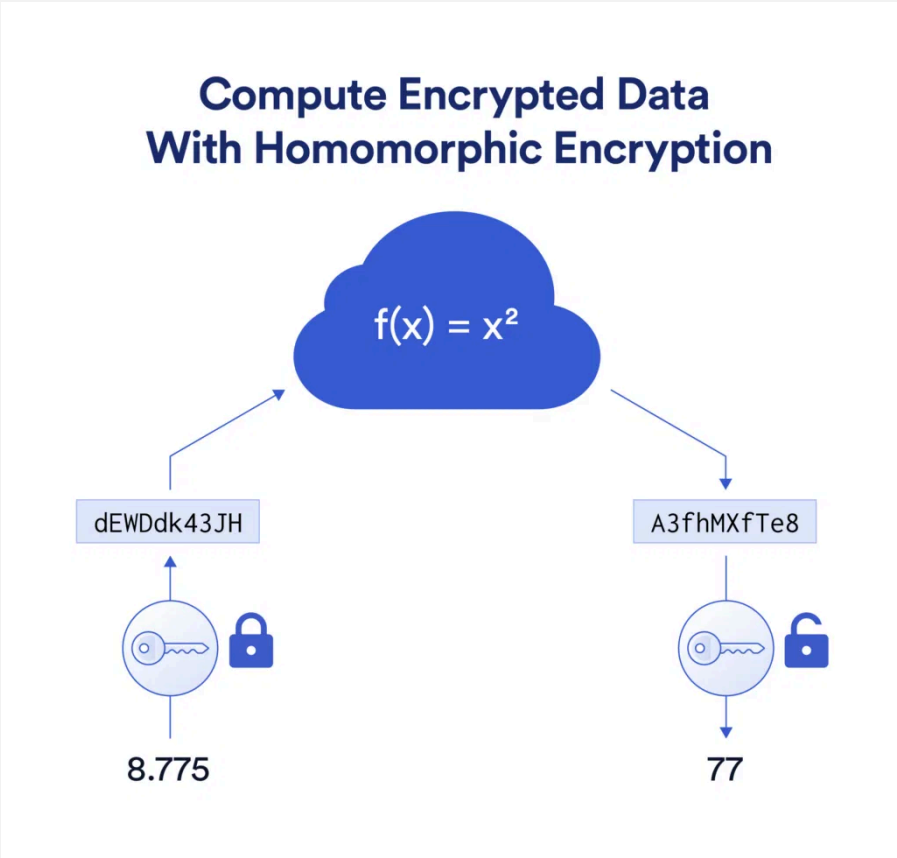


Performance Comparison: Orders of Magnitude Faster

MANA (QMNF) outperforms established FHE libraries across all core homomorphic operations.

Library	Homo Add	Homo Mul
MANA (QMNF)	2.85 μ s	21.91 μ s
Microsoft SEAL	~40 μ s	~5.5 ms
OpenFHE	~55 μ s	~6.2 ms
Zama TFHE-rs	109 ms	402 ms

Real-Time Capability: MANA's microsecond-level latency transforms FHE from a theoretical concept into a practical tool for real-time applications in privacy-preserving ML, secure databases, and financial processing.



The Impact: A New Era for Secure Computation

BOOTSTRAP-FREE RESCALING

K-Elimination eliminates the most expensive FHE operation, enabling sub-millisecond homomorphic operations.

LINEAR SCALABILITY

$O(k)$ complexity ensures performance remains high as RNS channels increase, enabling massively parallel secure computation.

HIGHEST ASSURANCE

Machine-checked proofs in Lean 4 and Coq provide formal verification, essential for sensitive data processing.

READY FOR ADOPTION

Seeking early adopters for privacy-preserving ML, secure MPC, encrypted databases, and financial data processing.

